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th National High School Physics Competition: Final Problems

I. (points) Fill in the blanks.

A uniform thin rubber ring has mass , stiffness and natural radius . Thrown aloft, it stays circular in steady motion. Ignore gravity. It translates and rotates uniformly at about its central normal axis. Its radius is .

A pigeon's whistle has frequency . Its top flight speed is . The Doppler effect makes observed frequencies range from to . (Sound travels at .)

A vertical insulated cylinder has mass and inner area , with a fixed amount of ideal gas at , sealed by a piston. Cyl.
The piston is insulated too. Neglect its mass and friction. Atmospheric pressure is ; gravity is . Pull the piston up slowly. At the opening, the cylinder just leaves the ground. Piston
In a slow adiabatic change is fixed; is pressure, is volume. is constant. At the opening, the gas temperature is .

Use two significant digits. Electron microscopes replace light with electrons to image objects. To resolve (atomic size), the wavelength cannot be larger. Thus the minimum electron speed is . To resolve structures, the minimum speed is . The required accelerating voltage is . The electron has rest mass and charge . Planck's constant is ; the speed of light is .

II. (points) As shown, a point-mass block slides from a platform down a slope onto a long horizontal belt at speed that carries it far away. The incline's height is , horizontal run , belt width , speed . Friction is . The block starts from rest at the top; while sliding, its velocity is normal to belt motion. A tiny circular arc joins the incline and belt, preserving its speed at the junction. The gravitational acceleration is .

What is the minimum block-belt friction coefficient to keep the block from sliding off the belt?

A DC motor with a speed regulator drives the belt. The connected supply has emf